School Worksheet

Class 12 Mathematics, Chapter 12: Linear Programming

Maximum marks: **27** **10** questions

Name _____ Date _____ Score _____ / 27

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

4 marks

1 MCQ — 1 mark

The corner points of the feasible region of an LPP are $(0, 2)$, $(3, 0)$, $(6, 0)$, $(6, 8)$ and $(0, 5)$. If $Z = 4x + 6y$, then the minimum value of Z occurs at:

(A) $(0, 2)$ only

(B) $(3, 0)$ only

(C) the mid-point of the line segment joining $(0, 2)$ and $(3, 0)$ only

(D) any point on the line segment joining $(0, 2)$ and $(3, 0)$

2 MCQ — 1 mark

The region represented by the system of inequations $3x + y \geq 3$, $2x - y \geq -5$, $x, y \geq 0$ is:

(A) unbounded in the 1st quadrant **(B)** bounded in the 1st quadrant

(C) unbounded in the 2nd quadrant **(D)** bounded in the 2nd quadrant

3 MCQ — 1 mark

The restrictions imposed on the decision variables involved in the objective function of a linear programming problem are called:

(A) feasible solutions **(B)** constraints

(C) optimal solutions **(D)** infeasible solutions

4 Assertion-Reason — 1 mark

Assertion (A): If the feasible region of an LPP is bounded, the objective function $Z = ax + by$ has both a maximum and a minimum value on it. **Reason (R):** If the feasible region is bounded, the maximum and minimum values of Z each occur at a corner point of the feasible region.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Q No.	Question	Marks
Level 2		**9 marks**
5	Short answer Solve the following linear programming problem graphically: Minimise $Z = 600x + 400y$ subject to the constraints $x + y \geq 8$, $x + 2y \leq 16$, $4x + y \leq 29$, $x, y \geq 0$.	3 marks
6	Short answer Solve the following LPP graphically: Minimise $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.	3 marks
7	Short answer Solve the following LPP graphically: Maximise $Z = 5x + 3y$ subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$.	3 marks
Level 3		**14 marks**
8	Long answer Solve graphically: Minimise and maximise $Z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.	5 marks
9	Long answer A furniture dealer deals in only two items, tables and chairs. He has ₹50,000 to invest and storage space for at most 60 pieces. A table costs ₹2,500 and a chair ₹500. He expects a profit of ₹250 on a table and ₹75 on a chair. Formulate this as an LPP and solve it graphically to find how many tables and chairs he should buy to maximise his profit, assuming he can sell all the items he buys.	5 marks
10	Case study **Case Study:** A dietician wants to prepare a mixture of two foods F_1 and F_2. One kg of F_1 contains 2 units of vitamin A and 1 unit of minerals; one kg of F_2 contains 1 unit of vitamin A and 2 units of minerals. The mixture must contain at least 8 units of vitamin A and 10 units of minerals. F_1 costs ₹50 per kg and F_2 costs ₹70 per kg. Let x kg of F_1 and y kg of F_2 be used. **(i)** Write the objective function and the constraints of the LPP. — 1 mark **(ii)** Find the corner points of the feasible region. — 1 mark **(iii)** Find the minimum cost of the mixture, and justify why the minimum exists even though the feasible region is unbounded. — 2 marks	4 marks

Answer key

Chapter 12: Linear Programming, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(D) any point on the line segment joining $(0, 2)$ and $(3, 0)$
2	(A) unbounded in the 1st quadrant
3	(B) constraints
4	(A) Both A and R are true, and R is the correct explanation of A

Level 2

5	Minimum $Z = 3200$ at $x = 0,\ y = 8$
6	Minimum $Z = 7$ at $x = \frac{3}{2},\ y = \frac{1}{2}$
7	Maximum $Z = \frac{235}{19}$ at $x = \frac{20}{19},\ y = \frac{45}{19}$

Level 3

8	Minimum $Z = 100$ at every point of the segment joining $(0, 50)$ and $(20, 40)$; maximum $Z = 400$ at $(0, 200)$
9	Buy 10 tables and 50 chairs; maximum profit ₹6,250
10	Minimum cost ₹380, using 2 kg of F_1 and 4 kg of F_2